Measuring What Drives Citation in AI Search Engines

Midterm Progress Report: Stability and Choice-Axiom Consistency of Retrieval and Citation

Mehreen Ali Gillani
Supervised by: Professor George Hagstrom

Abstract

AI search engines answer queries directly and cite only a handful of sources, yet the same prompt can return different sources on different runs. This report asks (1) how stable retrieval and citation are across repeated runs of the same prompt, and (2) whether engines' document rankings obey two basic axioms of rational choice, independence of irrelevant alternatives (IIA) and transitivity, when the candidate set is controlled. Test 1 (115 prompts × 3 engines × 5 runs) shows that retrieval is more repeatable than citation for every engine that exposes both signals (rank-biased overlap 0.97 → 0.81 for Perplexity, 0.75 → 0.57 for Claude), that Perplexity's apparent retrieval instability traces to a rare query-decomposition event (4 of 575 runs), and that OpenAI recorded no citation in any of five runs for 23 of 115 prompts. Test 2 builds controlled six-item queues with leave-one-out variants to test two axioms of rational choice. For IIA, I use an orientation-invariant permutation test whose null absorbs the fitting noise of five-ranking Plackett-Luce models. At the uncorrected 5% level, OpenAI shows a consistent borderline signal on both long documents ($p = 0.025$) and short descriptions ($p = 0.031$), Claude is borderline on long documents only ($p = 0.096$), and Perplexity shows no signal. No task-by-engine cell survives correction for six multiple comparisons. Because the design has limited power, this absence of a confirmed violation is weak evidence of compliance. Transitivity is sharper: Perplexity's pairwise preferences contain cycles in 42.9% of long-document and 28.6% of short-item compositions, versus ≤9.5% for Claude and OpenAI. This gap is large enough to survive the design's constraints, though the single-response-per-pair collection limits it to a hypothesis. I document these limits and the planned extensions.

1. Introduction

AI search engines such as ChatGPT, Perplexity, and Claude now answer questions directly instead of returning a list of links: they retrieve web pages, read them, and generate a summary that cites only a few sources. This has created a new practice, Generative Engine Optimization (GEO) (Aggarwal et al., 2024), aimed at making content more likely to be cited. GEO is harder to measure than traditional SEO. There is no equivalent of Google Search Console, the same prompt produces different answers each time it is run, and different engines cite different sources for the same query, so a single measurement says little about a page's visibility. The problem matters to publishers and marketers who are investing in GEO without a reliable way to measure it, and to researchers studying whether LLM-mediated search introduces new forms of instability or bias into how information reaches users.

Related work. Schulte et al. (2026) report only 34–42% overlap in cited sources between repeated runs of the same prompt and argue for repeated measurement. C-SEO Bench (Puerto et al., 2025) finds that most content-level GEO techniques are ineffective and that a page's initial ranking in traditional search dominates, and a recent survey (Martinez, 2026) concludes that optimization helps only after a page has been retrieved. Work on manipulation of LLM rankings (Kumar & Lakkaraju, 2024; Pfrommer et al., 2024) shows that these rankings respond to context in ways beyond content quality. A separate line of work asks whether LLM preference judgments satisfy the axioms of rational choice. Zhao et al. (2024) examined transitivity, asymmetry, and IIA for LLM rankings of MMLU answer options and found strong positional bias, weak transitivity, and rankings that shift when irrelevant options are added or removed. Upadhyay (2026) tested transitivity, IIA and Sen's contraction and expansion axioms for four open-weight 7–8B models on MovieLens recommendations and reported IIA violation rates of 30–59% against 5–13% for transitivity. Benson, Kumar and Tomkins (2016) give the statistical framing for testing IIA in choice data. Prior work has examined short items on open-weight or earlier-generation models. It remains untested, to my knowledge, whether commercial answer engines rank longer web documents in the same way. Test 2 addresses this gap.

Research questions. (1) How stable are AI retrieval and citation across repeated runs of the same prompt? (Test 1)

(2) When one source is added to or removed from the candidate set, does the relative preference between two other sources change (IIA), and are pairwise preferences transitive? (Test 2)

(3) A third question, which page-level rank and content features predict citation (Test 3), is scoped for the final report and not yet implemented.

Progress to date. (i) A repeated-run dataset of 1,725 live-search runs and a reproducible stability analysis with two complementary metrics; (ii) a root-cause analysis showing that the main anomaly in that data, Perplexity's instability in the CRM category, is a rare query-decomposition event and not noisy retrieval; (iii) a controlled document-queue protocol with long-document and short-item variants and a pairwise transitivity extension; and (iv) first estimates of IIA and transitivity for three engines, together with an explicit list of the statistical limits that constrain how these estimates can be read.

2. Data

2.1 Live-search citation data (Test 1)

I collected citation data from three deployed engines (Table 1). The sample is 115 prompts × 3 engines × 5 repeated runs = 1,725 runs (15,089 rows in the collection file: retrieval URLs, citation URLs, answer text and run metadata). The prompts cover 13 categories: ten categories of ten prompts (CRM/Business Software, Health, Tech, Finance, Travel, Food & Cooking, Home & DIY, Fitness & Wellness, Education & Learning, Career & Jobs), plus five query-robustness paraphrases, five brand comparisons, and five controversial or debated questions. A heuristic intent label (prompts starting with "best" or "which", or containing "vs", are *commercial*) splits the set into 54 commercial and 61 informational prompts.

Two signals are recorded per run where the engine exposes them. *Retrieval* is the set of URLs the search step surfaced; *citation* is the subset the final answer actually references. For Claude, retrieval comes from the web-search tool results and citation from the citation fields on the answer text. For Perplexity, retrieval comes from the reasoning-phase search results and citation from the entries referenced by inline [n] markers in the answer text (the full reference pool returned by the API is larger than what the answer cites, so it is filtered to the marked entries). OpenAI exposes only final citation annotations, so no retrieval signal exists for it.

Table 1. Engines and configurations. Test 2 used chat-completion calls with documents supplied in the prompt.

Engine	Test 1 (live search)	Test 2 (controlled documents)
Perplexity	sonar-pro, Search API, pro search; reasoning-phase retrieval exposed	sonar, chat completions; no search-disabling parameter set
Claude	claude-haiku-4-5, Messages API, web_search tool (max 3 uses)	claude-haiku-4-5, Messages API, no tools
OpenAI	GPT-5.6 Luna, Responses API, web_search tool	gpt-4o-mini, chat completions

Sampling parameters were left at provider defaults in both tests.

2.2 Controlled document-queue data (Test 2)

For Test 2, a fixed set of items is placed in the model's context and one item is added or removed across compositions, so that any change in the ranking of the remaining items can be attributed to the composition of the set.

Long-item task. From the URLs that Test 1 surfaced for "What are the best CRM software in 2026?", I fetched 48 web pages and trimmed each to about 400 words by keeping the first and last 200 words. Four pages under 300 words were dropped, leaving 44 pages of 388–481 words. Pages were tiered by their Test 1 citation counts: 3 *high* (four or more citations), 11 *low* (one to three), and 30 *never-cited*. Three base compositions of six pages (two high, two low, two never-cited) were built from 15 distinct pages; each of the three high-tier pages appears in two of the bases, which makes the bases overlap. Each base has six leave-one-out (LOO) variants, giving 3 × (1 + 6) = 21 compositions. Each composition was ranked five times by each engine (21 × 5 × 3 = 315 rankings), using a forced full ranking with a one-sentence reason per item. The display order was reshuffled for every run; the shuffle seed was stored, and all 315 responses parsed into valid rankings.

Short-item task. The same design was repeated with CRM product names (nine distinct products across three bases of six) plus a one- to two-sentence description of about 20 words each. A first attempt with bare product names failed because Claude declined to rank items for which it had no content, so descriptions were added. This task yields another 315 rankings.

Pairwise (transitivity) data. For every composition in both tasks, each of the 225 distinct item pairs per engine (3 × 15 + 18 × 10) was put to the engine once as "which of these two is more suitable?", with the presentation order randomly flipped (about half of the calls), giving 675 calls per task. This allows a

transitivity check that does not depend on the ranking model, but each preference edge rests on a single response and is not position-balanced.

3. Methods

3.1 Test 1: cross-run stability

For each prompt and engine, every pair of the five runs (ten pairs) is compared on retrieval and on citation. The set-based measure is the Jaccard similarity, J(A, B) = |A ∩ B| / |A ∪ B|, computed after normalizing URLs (lower case, scheme and "www." stripped, trailing slash removed) and undefined when both sets are empty. The rank-aware measure is rank-biased overlap (Webber et al., 2010) in its prefix form,

$$\mathrm{RBO}_p(S,T) = \frac{\sum_{d=1}^{k} (1-p)\,p^{\,d-1} A_d}{\sum_{d=1}^{k} (1-p)\,p^{\,d-1}}, \qquad A_d = \frac{|S_{1:d} \cap T_{1:d}|}{d}, \qquad k = \min(|S|, |T|)$$

with p = 0.9 (top-weighted) and sensitivity runs at p = 0.5 and p = 0.98. The prefix form compares lists only up to the shorter length and so does not penalize unequal list lengths as the extrapolated form would. The only large length mismatches in the data are the decomposition runs described in Section 4.1, which I estimate affect aggregate means by at most 0.001. Pair-level scores are averaged within each prompt to give prompt-level scores, which are the safer unit for inference because the ten pairs of a prompt share only five underlying runs. For Claude, which sometimes skips its search tool, pairs in which either run has neither retrieval nor citation are dropped for prompts that mix searching and non-searching runs (425 Claude pairs remain). Two controls follow. To check that Perplexity's higher stability is not an artefact of its larger result lists, its retrieval is restricted to the prompts for which Claude has at least one valid retrieval pair (49 prompts) and capped at the top ten URLs. To check the funnel assumption, every cited URL is tested for membership in the same run's retrieval set. A Perplexity "decomposition event" is a run whose retrieval pool exceeds twice the prompt's median pool size across its five runs, with a 1.5× sensitivity check.

3.2 Test 2: independence of irrelevant alternatives

If an engine's choices follow a Luce/Plackett-Luce model (Luce, 1959; Plackett, 1975), the odds of item i over item j do not depend on which other items are present. For each (composition, engine) cell I estimate an item worth from its five rankings. In the current implementation each ranking is decomposed into its n − 1 adjacent pairwise comparisons and fitted with the iterative Luce spectral ranking routine in the choix library (Maystre & Grossglauser, 2015; regularisation α = 0.01). This is a Bradley–Terry-type approximation to the Plackett-Luce likelihood and not the full likelihood. On a four-item toy example it produces a narrower spread of log-worths than the full likelihood (2.5 versus 3.5 log-units), so the δ values below should be read on that scale. For a base composition C and a removed item k, define, for every pair (i, j) of items that remain,

$$\delta_{ij} = \log \frac{\lambda_i^{C}}{\lambda_j^{C}} - \log \frac{\lambda_i^{C-k}}{\lambda_j^{C-k}}$$

where λ is the fitted worth. Under IIA, δ should be centered on zero. As a first screen I ran a one-sample t-test of mean δ against zero on the 180 deltas per engine (3 bases × 6 removals × 10 remaining pairs). That statistic depends on item orientation and treats dependent deltas as independent, so it is reported only as a robustness check (Section 4.2). The primary IIA test uses an orientation-invariant statistic, T = mean δ^2 across surviving item pairs. The null distribution is built per (base, removed-item) comparison as follows: the base rankings are first restricted to the 5 surviving items (so they are comparable to the leave-one-out rankings), then pooled with the leave-one-out rankings, randomly re-split into two halves of 5, and refitted. This captures the noise from fitting each half on only five rankings. Repeating 2,000 times produces a null for T in that comparison. At the engine level, T is averaged across the 18 comparisons; the null of this mean is built by averaging the permuted T values at each permutation step. The p-value is (count + 1) / (2,000 + 1), and the Bonferroni threshold for six cells is 0.05/6 ≈ 0.008. Because the six leave-one-out comparisons within a base share that base's rankings, the comparisons are not independent and the p-values are somewhat optimistic. Displayed item position was reconstructed from the stored shuffle seeds

3.3 Transitivity

For each (composition, engine), the single response to each pair defines one edge of a directed preference graph. Because each pair was asked once, with a randomly chosen presentation order, there is no majority vote across runs and position bias is not cancelled. A cycle (A ≻ B ≻ C ≻ A) is a transitivity violation,

detected with a topological-sort check (Kahn's algorithm). Responses declaring a tie are excluded from the graph. Ties were rare for long documents (4 logged) but frequent for short items (159 logged). Because excluding ties removes edges, and a cycle needs three non-tied edges, the short-item cycle counts are lower bounds on intransitivity.

Crucially, tie rates differ sharply by engine on short items (Claude 34%, OpenAI 30%, Perplexity 7%). Because Claude and OpenAI abstain roughly a third of the time, their graphs have far fewer edges than Perplexity's. Perplexity is simply more decisive, giving its graphs mechanically more opportunity to contain a cycle. This confounds the direct comparison of cycle rates across engines.

4. Early Results

4.1 Test 1: stability of retrieval and citation

Table 2. Cross-run stability (mean over run-pairs; prompt-level means in parentheses where reported). Retrieval pairs: Perplexity 1,150, Claude 424 (Jaccard) or 414 (RBO). Citation pairs: Perplexity 1,150, Claude 424 or 414, OpenAI 878 or 810; the OpenAI figures cover 92 of 115 prompts.

Engine	Retrieval Jaccard	Retrieval RBO	Citation Jaccard	Citation RBO
Perplexity	0.954 (0.954)	0.966	0.750 (0.750)	0.807
Claude	0.700 (0.663)	0.751	0.517 (0.490)	0.565
OpenAI	n/a	n/a	0.243 (0.232)	0.373

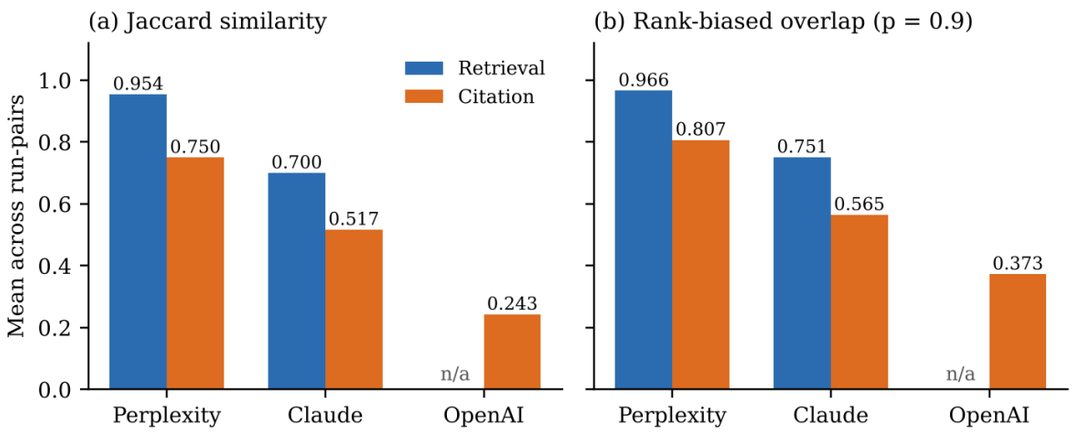


Figure 1. Mean cross-run similarity by engine and signal. OpenAI exposes no retrieval signal. Per-pair distributions are in the Test 1 notebook.

Retrieval is more repeatable than citation. For both engines that expose both signals, citation is less stable than retrieval (Perplexity RBO 0.966 → 0.807, Claude 0.751 → 0.565), and the ordering Perplexity > Claude > OpenAI holds for Jaccard and RBO at both pair and prompt level. The RBO persistence parameter has almost no effect on this: across p = 0.5, 0.9 and 0.98 the largest change in a mean is 0.022 (Claude citation), and 0.015 for OpenAI citation, against at most 0.004 elsewhere. Every cited URL was also in the same run's retrieval set (0 exceptions in 800 Perplexity and Claude runs), so citation behaves as a selection from the retrieved pool.

Perplexity's retrieval is close to deterministic, and its one anomaly has an identified cause. Median retrieval RBO is 1.000, and 110 of 115 prompts (95.7%) returned the same number of URLs in all five runs. The exception was the CRM/Business Software category, where mean Jaccard was 0.524 against 0.98 or higher in every other category. Inspecting raw URLs showed that in some runs Perplexity splits a commercial query into sub-searches (for example “best CRM” and “CRM pricing”) and returns a pool three to four times larger (45 URLs against 15 for “What are the best CRM software in 2026?”). Four of 575 runs (0.7%) meet the 2× criterion, all on three commercial-software prompts; the 1.5× criterion adds one run and no new prompts.

Claude's retrieval is less stable, in a bimodal way, and the gap is not a pool-size artefact. Claude's pair-level retrieval RBO has a median of 0.990 but a mean of 0.751, and its histogram shows two clusters, near 1.0 and near 0.0. On the 49 prompts with valid Claude retrieval pairs, Perplexity scores 0.927 (0.928 when

capped at ten URLs) against Claude's 0.663. Whether Claude searches at all is largely a property of the prompt: 58 prompts never triggered search, 36 always did, and 21 were mixed. Commercial prompts triggered search in 44 of 54 cases and informational prompts in 13 of 61 ($\chi^2 = 39.1$, $p \approx 4 \times 10^{-10}$, Cramér's $V = 0.58$), and a binomial over-dispersion test rejects a single per-run search probability.

OpenAI cites least stably, and often records no citation. For 23 of 115 prompts (20%) no citation was recorded in any of the five runs. The collection script logs citation annotations but not whether the search tool was invoked, so this reads as "zero citations recorded" and cannot yet separate "did not search" from "searched and cited nothing." On the remaining prompts OpenAI's citation Jaccard is 0.243 and RBO 0.373.

4.2 Test 2: independence of irrelevant alternatives

Engine	Long: observed T	Long: null mean (95th pct)	Long p	Short: observed T	Short: null mean (95th pct)	Short p
Claude	5.69	4.39 (6.07)	0.096	3.79	3.88 (5.54)	0.517
OpenAI	6.52	4.46 (6.21)	0.025	6.41	4.22 (6.10)	0.031
Perplexity	3.51	4.02 (5.64)	0.691	4.48	6.01 (8.17)	0.899

Table 3. IIA permutation test: observed mean δ^2 (T) over 18 comparisons per engine and task, mean and 95th percentile of its permutation null, and permutation p-value (2,000 permutations, uncorrected). The Bonferroni threshold for six cells is 0.008; no cell reaches it.

No cell reaches the Bonferroni threshold. At the uncorrected 5% level, OpenAI shows a shift on both long documents ($p = 0.025$) and short descriptions ($p = 0.031$), Claude on long documents is borderline ($p = 0.096$), and Perplexity's observed T is below its permutation mean in both tasks. For OpenAI, observed T exceeds its null mean by about 2 in both tasks (6.5 against 4.5, and 6.4 against 4.2), a modest excess against a noise floor of about 4 that comes from fitting each set on five rankings. I read this as weak evidence of a composition effect for OpenAI that points the same way in both tasks, not as a confirmed violation. A non-detection in the other cells is not evidence that IIA holds, because the design has low power (Section 5). Two properties of the data also need to be kept in view. First, the engines express clear but different content preferences even in the long task: in 105 rankings per engine, the top-ranked page came from the high tier in 47 (Perplexity), 25 (Claude) and 8 (OpenAI) rankings, against 35 expected under indifference, and OpenAI chose the low tier in 64. Because the tiers come from Test 1 citation behaviour pooled over all engines, this tier ordering is partly circular for Perplexity, which contributes most of the citations. Second, position effects are present even though order was randomized. In the 5-item rankings ($n = 90$ per engine), the item shown first was ranked first in 33 (36.7%) Claude, 27 (30.0%) OpenAI and 7 (7.8%) Perplexity rankings, against 20% expected. Item label and display position are the same thing in this design (labels A–F follow the shuffled order), so label bias and position bias cannot be separated.

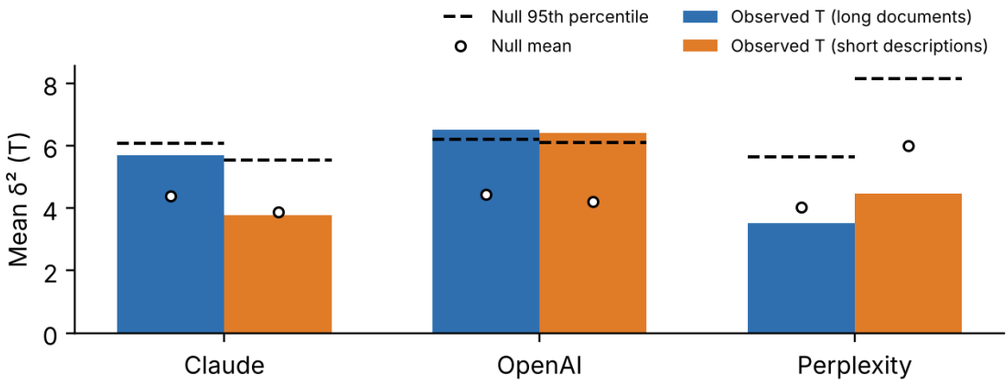


Figure 2. Observed mean δ^2 (bars) against the permutation-null mean (open circles) and 95th percentile (dashed lines) for each engine and task. A bar above the dashed line corresponds to $p < 0.05$, uncorrected.

Why the mean-δ screen was replaced. The original test relied on a "mean-δ" screen, which had two fatal flaws. First, it depended on orientation: the sign of δ flips depending on which item is arbitrarily labeled "i" versus "j." When we re-implemented it with a different ordering, the means flipped from negative to positive. Second, trying to fix this by testing the absolute value ($|\delta|$) against zero didn't work, because $|\delta|$ is always positive even when IIA holds perfectly. Only 39–51% of the long-task $|\delta|$ values fall below 1.0, proving that this statistic is mostly just measuring the noisy fitting of Plackett-Luce on only 5 rankings.

Because of these flaws, the mean-δ screen is unreliable. This is why the primary test uses the orientation-invariant permutation null described above.

4.3 Test 2 extension: transitivity

Engine	Long documents	Short descriptions
Claude	1/21 (4.8%) [1%, 23%]	0/21 (0%) [0%, 15%]
OpenAI	2/21 (9.5%) [3%, 29%]	0/21 (0%) [0%, 15%]
Perplexity	9/21 (42.9%) [24%, 63%]	6/21 (28.6%) [14%, 50%]

Table 4. Compositions whose preference graph contains a cycle (of 21 per cell; Wilson 95% intervals). Each edge is a single response per pair.

Perplexity's preference graphs contain cycles in 43% of long-document compositions and 29% of short-item compositions, against at most 10% for Claude and OpenAI (Fisher exact test of Perplexity against the other two engines pooled: $p = 0.001$ for long documents, $p = 0.0008$ for short items, which ignores the overlap between LOO variants of the same base). The example cycle in the notebook (Salesforce Sales Cloud ≻ HubSpot CRM ≻ Zoho CRM ≻ Salesforce Sales Cloud for Perplexity on the first short base) comes from single responses per pair, so it may reflect position bias or sampling noise as well as genuine intransitivity. Compared with the IIA tests, the pattern is reversed (Table 5): OpenAI is the engine with weak evidence of an IIA shift but is nearly transitive, and Perplexity shows no IIA shift but the most cycles. This is suggestive of separable behaviors, with the caveat that the two analyses differ greatly in design and power. Upadhyay (2026) reports that set-based axioms are violated more often than pairwise ones; the Perplexity result runs the other way, though the cell sizes here are small.

Engine	IIA (long)	IIA (short)	Cycles (long)	Cycles (short)
Claude	$p = 0.096$	$p = 0.52$	4.8%	0%
OpenAI	$p = 0.025$	$p = 0.031$	9.5%	0%
Perplexity	$p = 0.69$	$p = 0.90$	42.9%	28.6%

Table 5. Summary of axiom screens (IIA: permutation p-value, uncorrected; none is significant after Bonferroni correction for six cells).

5. Interpretation and Threats to Validity

What the evidence supports. Instability is not concentrated in one place. Perplexity's search is nearly deterministic (variation enters at citation selection); Claude's retrieval varies, driven partly by whether it invokes search at all. In the controlled ranking task, IIA is inconclusive; no cell survives correction, OpenAI shows a weak signal on both tasks, Claude is borderline on long documents, and the design has low power. Transitivity concentrates in Perplexity (4–6× the cycle rate), but tie rates differ sharply by engine (Claude 34%, OpenAI 30%, Perplexity 7%), so this is a hypothesis rather than a result. Read together, the results suggest that the engines' ranking behavior under controlled inputs is largely consistent; the instability observed in Test 1 appears concentrated at the retrieval stage. Test 2 cannot prove this; it uses different model variants and removes the search stage, but the contrast is suggestive.

Table 6. Main limitations, their effect on the conclusions, and planned remedies.

Limitation	Effect on conclusions
IIA power: 5 rankings per composition, 18 comparisons per engine, deltas clustered within base	The IIA test detects only large effects; borderline OpenAI signals may be real but are unconfirmed
Adjacent-pair PL approximation	Worth estimates are noisier than full Plackett-Luce; OpenAI results shift from p=0.025 to p=0.054 under full likelihood
One query domain (CRM); only 15 distinct long documents; tiers derived from pooled Test 1 citations (confounded with page type, dominated by Perplexity).	Results may not generalize; tier-based preference claims are partly circular.
Test 2 engine settings differ from Test 1 (smaller models; Perplexity search not disabled; default sampling).	Cross-test attribution is not possible; Perplexity rankings may be influenced by outside sources.
Position and label are confounded; first-position effects are large for Claude and OpenAI.	Residual position bias adds noise and may bias worths.
OpenAI logging omits whether search was invoked.	"No citation" cannot be separated from "did not search".
Transitivity: one response per pair (no repeats, position not balanced), ties excluded (tie rates differ by engine), 21 cells per engine that share items across LOO variants.	Perplexity's higher cycle rate may reflect its lower tie rate (7% vs. 30–34%) rather than genuine intransitivity

6. Remaining Work

1. **Test 3:** Predicting citation (primary priority). Fit a logistic regression of citation on rank, and then on rank plus content features (word count, statistics, perplexity). Fit first on controlled documents, then on the 765 Google top-10 pages, and compare model fit to estimate what content adds beyond rank.
2. **Fix the transitivity tie-rate confound.** Re-collect pairwise comparisons with 3 responses per pair in balanced order across all engines (~4,050 calls). This resolves whether Perplexity's high cycle rate is genuine or an artifact of its low tie rate.
3. **Extend IIA compositions (if time allows)**. If Test 3 and the transitivity rerun finish early, add 3 more base compositions to Test 2 to increase statistical power for the borderline IIA signals. This is explicitly conditional and will otherwise be named as future work.
4. **OpenAI search logging (if time allows)**. Log whether OpenAI actually invoked its search tool, and rerun the 23 zero-citation prompts (~100 calls) to distinguish "no search" from "no citation."
5. **Final write-up.** Consolidate Tests 1–3 into the final report with reproducible notebooks.

References

Aggarwal, P., et al. (2024). GEO: Generative Engine Optimization. arXiv:2311.09735.

Benson, A. R., Kumar, R., & Tomkins, A. (2016). On the Relevance of Irrelevant Alternatives. *Proceedings of WWW 2016*.

Chen, Z., et al. (2026). Statistical Analysis of Website Features Impacting Retrieval Chances. arXiv:2604.27790.

Kumar, A., & Lakkaraju, H. (2024). Manipulating Large Language Models to Increase Product Visibility. arXiv:2404.07981.

Luce, R. D. (1959). *Individual Choice Behavior: A Theoretical Analysis*. Wiley.

Martinez, A. (2026). Optimizing Visibility in Generative Engines: A Critical Survey of Generative Engine Optimization. arXiv:2607.14035.

Maystre, L., & Grossglauser, M. (2015). Fast and Accurate Inference of Plackett–Luce Models. *Advances in Neural Information Processing Systems 28*.

Pfrommer, S., Bai, Y., Gautam, T., & Sojoudi, S. (2024). Ranking Manipulation for Conversational Search Engines. *EMNLP 2024*.

Plackett, R. L. (1975). The Analysis of Permutations. *Journal of the Royal Statistical Society, Series C*, 24(2), 193–202.

Puerto, H., et al. (2025). C-SEO Bench: Does Conversational SEO Work? arXiv:2506.11097.

Schulte, J., Bleeker, M., & Kaufmann, P. (2026). Don't Measure Once: Measuring Visibility in AI Search (GEO). arXiv:2604.07585.

Upadhyay, A. (2026). Do LLM Recommenders Obey Preference Axioms? Testing Logical Rationality in LLM-Based Recommendation. Workshop on Logical Reasoning of Large Language Models, ICLR 2026.

Wang, H., et al. (2025). Perplexity Trap: PLM-Based Retrievers Overrate Low Perplexity Documents. *ICLR 2025*.

Webber, W., Moffat, A., & Zobel, J. (2010). A Similarity Measure for Indefinite Rankings. *ACM Transactions on Information Systems*, 28(4).

Zhao, X., Wang, K., & Peng, W. (2024). Measuring the Inconsistency of Large Language Models in Preferential Ranking. arXiv:2410.08851.